The SilkMind Caravan

Test Bank V1.0 · Software & Symbolic AI

Team SilkMind · Stage 2 · November 2026 · Companion to the Study Guide V1.0 (built from Markdown Content V2.0)

How to use this test bank. Questions come first; the answer key with explanations starts at Part D. Every question shows the guide section it tests in [brackets], e.g. [S2 · Mission 2.1] = Stop 2, Mission 2.1. All data needed (rules, tables) is on the Reference Sheet, so no website is needed. Multiple-choice questions say whether to choose ONE answer or ALL that apply.

Part	Questions	What it checks
A · True / False	24	Meaningful understanding; false statements are corrected in the key
B · Multiple choice	24	Applying ideas: traces, predictions, finding errors
C · Open-ended	8	Explaining, comparing, evaluating (aim for 120–250 words or worked steps)
D · Answer key	–	Answers, explanations, expected key points and sample answers

Coverage

Guide stop (learning outcomes)	True/False	Multiple choice	Open-ended
Stop 1 · Software & logic (LO1–2)	6	5	1
Stop 2 · Rules & inference, Labs 2A/2B (LO3–4)	8	9	3
Stop 3 · Machine learning, Lab 3 (LO5)	4	6	2
Stop 4 · History & AI winters (LO6)	4	2	1
Stop 5 · Modern software, Lab 5 (LO7)	2	2	1

Reference Sheet

Use these exactly as in the Study Guide.

Rules R1–R6 (Stop 2)

Rule	IF …	THEN …
R1	has feathers	is a bird
R2	gives milk	is a mammal
R3	is a bird AND cannot fly AND swims	is a penguin
R4	is a bird AND can fly AND hunts at night	is an owl
R5	is a mammal AND eats meat AND has black stripes	is a tiger
R6	is a mammal AND has hooves AND has black stripes	is a zebra

Forward chaining: in each cycle, check R1→R6 in order and fire the first rule whose IF part is completely in working memory and whose THEN fact is new; stop when a cycle adds nothing.

Training messages (Stop 3, Mission 3.1)

Message	A	B	C	D	E	F	G	H	I	J
Links	0	1	1	2	3	2	4	5	6	8
Label	not	not	not	not	not	scam	scam	scam	scam	scam

Model shape: IF links ≥ k THEN scam ELSE not. Extra message for retraining: N, 3 links, scam.

Guard rules G1–G4 (Stop 5)

Rule	IF current state is …	AND …	THEN new state
G1	Wander	player is seen	Attack
G2	Attack	health is low	Flee
G3	Attack	no player is present	Wander
G4	Flee	no player is present	Wander

If no rule matches, the character stays in its current state.

Part A · True / False

Write T or F. In the key, every false statement is corrected.

TF-01 An operating system, such as Android, is an example of hardware. *[S1 · Mission 1.2 (Real thing)]* **T / F**

TF-02 An algorithm must finish after a finite number of clear steps. *[S1 · Real thing]* **T / F**

TF-03 The expression A OR B is false when both A and B are true. *[S1 · Real thing]* **T / F**

TF-04 With the rule IF T < 18 THEN heater ON ELSE heater OFF, the heater is ON when the temperature is exactly 18 °C. *[S1 · Mission 1.3]* **T / F**

TF-05 The login rule access = account_exists OR password_matches would let someone in who types a real user's email with a wrong password. *[S1 · Mission 1.2 Part B]* **T / F**

TF-06 Pseudocode can be run directly by a computer. *[S1 · Traps]* **T / F**

TF-07 In a rule-based system, the knowledge base changes for every new case. *[S2 · Real thing (parts)]* **T / F**

TF-08 A rule whose IF part is joined by AND fires only when every one of its conditions is in working memory. *[S2 · Mission 2.1]* **T / F**

TF-09 In forward chaining, a fact added in one cycle can make another rule fire in the next cycle. *[S2 · Mission 2.1]* **T / F**

TF-10 Backward chaining starts from a goal and works back to the facts needed to prove it. *[S2 · Real thing (chaining)]* **T / F**

TF-11 If an expert system concludes that an animal is both a tiger and a zebra, the inference engine must be broken. *[S2 · Mission 2.1 last round]* **T / F**

TF-12 In Prolog, a word that starts with a capital letter, such as Pingu, is treated as a variable. *[S2 · Lab 2A]* **T / F**

TF-13 ELIZA understands the meaning of the sentences typed to it. *[S2 · Lab 2B]* **T / F**

TF-14 The MYCIN expert system could explain the reasoning behind its conclusions when asked. *[S2 · Real expert systems]* **T / F**

TF-15 In machine learning, labelled data means examples that come with the correct answer. *[S3 · Real thing (ML terms)]* **T / F**

TF-16 In Mission 3.1, the best rule learned from the 10 training messages (links ≥ 4) made no mistakes on those messages. *[S3 · Mission 3.1]* **T / F**

TF-17 Accuracy measured on the same examples that were used for training is a reliable estimate of accuracy on new examples. *[S3 · Mission 3.1 Reflection]* **T / F**

TF-18 A trained model can be thought of as a rule that was chosen by a training process rather than written by a person. *[S3 · Traps]* **T / F**

TF-19 An 'AI winter' is a period in which funding for and interest in AI research fall sharply. *[S4 · Real thing]* **T / F**

TF-20 The Lighthill report (1973) praised AI research for meeting its early promises. *[S4 · Real thing / primary source]* **T / F**

TF-21 Adding one more digit to a code lock (digits 0–9) multiplies the number of possible codes by ten. *[S4 · Mission 4.2]* **T / F**

TF-22 During the AI winters, all AI research and all expert systems stopped. *[S4 · Traps]* **T / F**

TF-23 A finite-state-machine game character stays in its current state when none of its transition rules match. *[S5 · Real thing (game characters)]* **T / F**

TF-24 In AlphaGeometry, the language model performs the step-by-step proof while the symbolic engine suggests new constructions. *[S5 · Real thing (neuro-symbolic)]* **T / F**

Part B · Multiple choice

MC-01 Which of these is software? **(Choose ONE answer.)** *[S1 · Real thing]*

a) a keyboard
b) a temperature sensor
c) a web browser
d) a memory chip

MC-02 has_ticket = True and ticket_expired = True. What is has_ticket AND NOT ticket_expired? **(Choose ONE answer.)** *[S1 · Checkpoint]*

a) True
b) False
c) Both true and false
d) It cannot be determined

MC-03 Which of these can be used as Boolean conditions (always exactly true or false)? **(Choose ALL that apply.)** *[S1 · Mission 1.1 / Real thing]*

a) T < 18
b) "it is a bit cold"
c) account_exists
d) "the password looks strong-ish"

MC-04 A fridge alarm uses IF T > 5 THEN alarm ON ELSE alarm OFF. What happens at 4, 5 and 6 °C? **(Choose ONE answer.)** *[S1 · Mission 1.3]*

a) OFF, ON, ON
b) OFF, OFF, ON
c) ON, ON, ON
d) OFF, OFF, OFF

MC-05 Why should a login page show the same message for 'unknown email' and 'wrong password'? **(Choose ONE answer.)** *[S1 · Mission 1.2 Reflection]*

a) It saves memory
b) So attackers cannot learn which email addresses have accounts
c) Because the program cannot tell the two cases apart
d) Because Boolean logic only allows one message

MC-06 Which part of a rule-based system repeatedly finds rules whose IF part is true and fires them? **(Choose ONE answer.)** *[S2 · Real thing (parts)]*

a) The knowledge base
b) Working memory
c) The inference engine
d) The user

MC-07 Starting facts: has feathers, can fly, hunts at night. What does forward chaining with R1–R6 conclude? **(Choose ONE answer.)** *[S2 · Mission 2.1 (rules R1–R6)]*

a) is a bird only
b) is a bird, then is an owl
c) is an owl only (R4 fires first)
d) is a penguin

MC-08 Starting facts: gives milk, swims. What does the system conclude with R1–R6? **(Choose ONE answer.)** *[S2 · Lab 2A / Mission 2.1]*

a) is a mammal only
b) is a mammal and is a penguin
c) nothing at all
d) is a zebra

MC-09 A student traces gives milk, has hooves, has black stripes as 'Cycle 1: R6 fires → is a zebra'. What is the error? **(Choose ONE answer.)** *[S2 · Mission 2.1]*

a) R6 should never fire for these facts
b) 'is a mammal' is not yet in working memory; R2 must fire first
c) The correct answer is tiger
d) Forward chaining always starts with the last rule

MC-10 The 'mystery' animal has: gives milk, eats meat, has hooves, has black stripes. Which statements are true? **(Choose ALL that apply.)** *[S2 · Lab 2A query 4]*

a) The system concludes 'tiger'
b) The system concludes 'zebra'
c) The inference engine has a bug
d) The knowledge base contains a rule conflict

MC-11 Goal: 'is an owl?'. The system knows only 'has feathers'. Using backward chaining, what must it ask the user? **(Choose ONE answer.)** *[S2 · Mission 2.2]*

a) Does it give milk?
b) Can it fly? and Does it hunt at night?
c) Does it swim? and Can it not fly?
d) Every possible question about the animal

MC-12 In the Prolog rule is_a(X, bird) :- fact(X, has_feathers). what does :- mean? **(Choose ONE answer.)** *[S2 · Lab 2A (Just enough Prolog)]*

a) AND
b) OR
c) IF
d) NOT

MC-13 A learner adds fact(gerald, has_long_nek). (a typo) but the giraffe rule needs has_long_neck. What does is_a(gerald, giraffe). return? **(Choose ONE answer.)** *[S2 · Lab 2A Build]*

a) true: Prolog corrects small spelling mistakes
b) false
c) An error message about spelling
d) Prolog asks the user what was meant

MC-14 ELIZA answered 'i am a robot made of cheese' with 'Do you know anyone else who is a robot made of cheese ?'. What is the best explanation? **(Choose ONE answer.)** *[S2 · Lab 2B]*

a) It understood the joke and played along
b) It matched a pattern and copied the user's words into a reply template
c) It searched the internet for robots made of cheese
d) It learned the reply from millions of conversations

MC-15 Using the ten training messages A–J, how many mistakes does the rule IF links ≥ 2 THEN scam make? **(Choose ONE answer.)** *[S3 · Mission 3.1 (data table)]*

a) 1
b) 2
c) 3
d) 0

MC-16 Which threshold k gives the fewest mistakes on the original ten messages? **(Choose ONE answer.)** *[S3 · Mission 3.1]*

a) k = 2
b) k = 3
c) k = 4
d) k = 5

MC-17 Message N (3 links, scam) is added and the model retrained. Which statement is true? **(Choose ONE answer.)** *[S3 · Mission 3.1 retrain]*

- **a)** k = 4 is still the only best rule
- **b)** k = 2, 3 and 4 now tie with 2 mistakes each
- **c)** k = 5 becomes the best rule
- **d)** Every k now makes at least 5 mistakes

MC-18 Which are advantages of machine learning compared with hand-written rules? **(Choose ALL that apply.)** *[S3 · Real thing / comparison table]*

- **a)** It can learn patterns too complex to write by hand
- **b)** Its answers are always easy to explain
- **c)** It can be retrained when the data changes
- **d)** It needs no data

MC-19 In Lab 3 the genuine message 'Free tea tasting at our shop on Friday' may be labelled 'scam'. What is the most likely reason? **(Choose ONE answer.)** *[S3 · Lab 3 Investigate]*

- **a)** The model understands that free tea is suspicious
- **b)** It shares words (e.g. 'free') with the scam training examples
- **c)** Machine Learning for Kids is broken
- **d)** There were too many genuine examples

MC-20 A bank must explain every refused loan using its published criteria. Which approach fits best? **(Choose ONE answer.)** *[S3 · Mission 3.2]*

- **a)** Machine learning only
- **b)** Symbolic (rule-based) AI
- **c)** Neither: decide at random
- **d)** A chatbot like ELIZA

MC-21 Which list is in the correct time order? **(Choose ONE answer.)** *[S4 · Mission 4.1]*

- **a)** Dartmouth workshop → Lighthill report → XCON in use → LISP-machine collapse
- **b)** Lighthill report → Dartmouth workshop → XCON in use → LISP-machine collapse
- **c)** Dartmouth workshop → XCON in use → Lighthill report → LISP-machine collapse
- **d)** XCON in use → Dartmouth workshop → LISP-machine collapse → Lighthill report

MC-22 Which contributed to the second AI winter (≈1987–1993)? **(Choose ALL that apply.)** *[S4 · Real thing (winters)]*

- **a)** Expert systems were brittle on cases their rules did not cover
- **b)** Large rule bases were costly to maintain
- **c)** The market for LISP machines collapsed
- **d)** The invention of Boolean logic

MC-23 Delivery fees: F1 less than 5 km €2; F2 more than 5 and at most 10 km €4; F3 more than 10 km €6. Which distance matches no rule? **(Choose ONE answer.)** *[S5 · Mission 5.3]*

- **a)** 4 km
- **b)** 5 km
- **c)** 10 km
- **d)** 11 km

MC-24 A guard following rules G1–G4 is in Wander. The player is seen and the guard's health is low. What is its new state? **(Choose ONE answer.)** *[S5 · Mission 5.2]*

- **a)** Flee
- **b)** Attack
- **c)** Wander
- **d)** An error

Part C · Open-ended

Answer in full sentences (normally 120–250 words), or with worked steps, tables or diagrams where asked. Use evidence from the guide's activities.

OE-1 A caravanserai thermostat uses IF T < 18 THEN heater ON ELSE heater OFF. A colleague wants to change it to T <= 18 because 'it's basically the same'. Explain, with a small truth table for 17, 18 and 19 °C, whether the two rules are the same. Then explain why this level of precision matters for software, using one real-world consequence. *[S1 · Mission 1.3]*

OE-2 Using rules R1–R6, trace forward chaining for the starting facts has feathers, cannot fly, swims. Present a trace table (cycle, facts at start, rule fired, new fact), explain why the engine stops, and write the 'why?' explanation chain the system could give. Finally, explain why such explanations made systems like MYCIN easier to trust. *[S2 · Missions 2.1, 2.3]*

OE-3 In Lab 2A, SWISH answered the query is_a(mystery, What). with What = mammal, then tiger, then zebra (facts: gives milk, eats meat, has hooves, has black stripes). Explain what happened and whether it is a fault of Prolog. Propose a change to the knowledge base that fixes it, explain why your change works, and describe how you would test that the fix did not break other cases. *[S2 · Lab 2A, Mission 2.1]*

OE-4 Here are three lines from our ELIZA test: (1) 'i think computers are scary' → 'What do you think about machines ?'; (2) 'the bus to samarkand leaves at nine' → 'Lets discuss further why your exams.' (earlier the user had typed 'i am worried about my exams'); (3) 'what is two plus two' → 'What do you think ?'. Explain how ELIZA's rules produced each reply, and use the examples to discuss what ELIZA does and does not demonstrate about intelligence. *[S2 · Lab 2B]*

OE-5 The merchants of Samarkand want a scam-message filter. Compare a rule-based filter (BEK) with a machine-learning filter (To'ti), using at least three criteria from the guide's comparison table and evidence from Mission 3.1 or Lab 3. Recommend an approach (or a combination) and justify it. *[S3 · Mission 3.1, Lab 3, Mission 3.2]*

OE-6 A classmate says: 'Our model IF links ≥ 4 THEN scam made only 1 mistake on the 10 training messages, so it will be 90% accurate on every future message.' Evaluate this claim. Explain the difference between training data and test data, give at least two reasons the claim is weak, and describe a better way to estimate accuracy. *[S3 · Mission 3.1 Reflection]*

OE-7 Explain why the second AI winter (≈1987–1993) happened. Your answer must include one 'expectations' cause, one 'technology' cause, and a link to at least two limits you observed yourself in Stop 2 activities. Use the idea of the combinatorial explosion to explain why these limits became worse as systems grew. *[S4 · Real thing, Missions 4.2, 4.4]*

OE-8 In Lab 5, the working rule inside(X, Z) :- located_in(X, Y), inside(Y, Z). correctly derived that the Registan is inside Central Asia. When the rule was changed to inside(X, Z) :- inside(X, Y), inside(Y, Z). the query inside(registan, mongolia). produced 'Stack limit exceeded … Probable infinite recursion'. Explain why the broken rule never stops, why the working rule does stop, and what this example shows about building and checking rule-based systems. *[S5 · Lab 5]*

Part D · Answer key

Checked by: answers verified against the Study Guide Feedback Corners, the SWISH/ELIZA test runs (3 Nov 2026) and the demo programs examples/topic02_mini_rule_engine.py and topic03_learn_a_rule.py. Peer review by a second team member: see the review record at the end.

A · True / False

#	Answer	Explanation / correction
TF-01	F	False. An operating system is (system) software: instructions. Hardware is the physical machine: screen, processor, sensors.
TF-02	T	True. That is the definition used in the guide (Britannica): finite, clear, ordered steps.
TF-03	F	False. Boolean OR is true when at least one part is true, including when both are true.
TF-04	F	False. 18 < 18 is false, so the ELSE part runs and the heater is OFF. 'Below 18' means strictly less than 18.
TF-05	T	True. account_exists is true, so OR is true. The correct rule uses AND.
TF-06	F	False. Pseudocode is plain-language planning of an algorithm; it is not a programming language.
TF-07	F	False. Working memory (the facts of this case) changes; the knowledge base (the rules) stays the same.
TF-08	T	True. AND needs all conditions to be true; e.g. R5 does not fire for 'gives milk, has hooves, has black stripes' because 'eats meat' is missing.
TF-09	T	True. E.g. R1 adds 'is a bird' in cycle 1; that lets R3 fire in cycle 2. This is chaining.
TF-10	T	True. It is goal-driven; forward chaining is data-driven.
TF-11	F	False. The engine did exactly what the rules say. The knowledge base has a rule conflict: R5 and R6 both accept a striped mammal.
TF-12	T	True. So is_a(Pingu, penguin). asks 'which animal is a penguin?' and answers Pingu = pingu.
TF-13	F	False. ELIZA finds keywords, matches a pattern and fills a reply template, swapping pronouns. It does not know what any word means (e.g. it reflected 'a robot made of cheese').
TF-14	T	True (Britannica). Explanation from named rules is a key strength of rule-based systems.
TF-15	T	True. E.g. each message card in Mission 3.1 carries the label scam / not.
TF-16	F	False. It made 1 mistake: message F (2 links, scam) is predicted 'not'. The best model is not perfect.
TF-17	F	False. The model was fitted to those examples, so it usually does worse on new ones. Test on new labelled examples that were not used in training.
TF-18	T	True. In Mission 3.1 the model literally is the rule IF links ≥ 4 THEN scam; the data chose the 4.
TF-19	T	True. Usually after expectations were set too high and results did not match.
TF-20	F	False. Lighthill concluded that 'in no part of the field have the discoveries made so far produced the major impact that was then promised'.
TF-21	T	True. 4 digits = 10,000 codes; 5 digits = 100,000. This rapid growth is the combinatorial explosion.
TF-22	F	False. Funding and interest fell sharply, but research continued on a smaller scale and some systems (e.g. XCON) stayed in use.
TF-23	T	True. E.g. a guard in Attack with full health and the player visible stays in Attack.
TF-24	F	False. It is the other way round: the symbolic deduction engine proves step by step; the language model suggests a construction when the engine is stuck.

B · Multiple choice

#	Answer	Reasoning
MC-01	c	A web browser is a program (application software). The others are physical hardware.

#	Answer	Reasoning
MC-02	b	NOT True = False, and True AND False = False. The traveller cannot use the ticket.
MC-03	a, c	T < 18 and account_exists are always true or false. 'A bit cold' and 'strong-ish' are vague: a computer cannot test them.
MC-04	b	Only 6 > 5 is true. At exactly 5, '5 > 5' is false, so the alarm is OFF: the same boundary trap as 18 °C.
MC-05	b	Different messages would reveal which emails are registered. OWASP recommends one generic message. The program can tell the cases apart (b is the reason, not c).
MC-06	c	The inference engine is the 'loop'. The knowledge base stores rules; working memory stores the case's facts.
MC-07	b	Cycle 1: R1 → is a bird. Cycle 2: R4 (bird AND can fly AND hunts at night) → is an owl. R4 cannot fire first because 'is a bird' is not yet known.
MC-08	a	R2 fires (is a mammal). No rule uses 'swims' with 'is a mammal', so it stops: a gap in the knowledge base.
MC-09	b	Correct trace: cycle 1 R2 → mammal; cycle 2 R6 → zebra; cycle 3 stop. A right answer does not prove a right trace.
MC-10	a, b, d	SWISH returned mammal, tiger, zebra. R5 and R6 overlap, a rule conflict in the knowledge base; the engine works correctly.
MC-11	b	Only R4 concludes owl: it needs is a bird (proved by R1 from has feathers), can fly and hunts at night. It asks only those two.
MC-12	c	Read ':-' as IF: X is a bird IF X has feathers. A comma means AND.
MC-13	b	To a computer the two symbols are simply different, so the condition is not met and Prolog answers false. It never guesses meanings.
MC-14	b	ELIZA (1966) uses keyword/pattern rules and templates. It repeats even nonsense, showing no understanding.
MC-15	b	D (2 links, not) and E (3 links, not) are wrongly called scam; F (2 links, scam) is correct. Total 2.
MC-16	c	Mistakes for k = 0…9 are 5, 4, 2, 2, 1, 2, 3, 4, 4, 5. k = 4 has just 1 mistake (message F).
MC-17	b	k = 2: D, E wrong; k = 3: E, F wrong; k = 4: F, N wrong, each 2 mistakes. One new example changed the model (checked with topic03_learn_a_rule.py).
MC-18	a, c	ML is flexible and retrainable, but needs plenty of good labelled data and is often hard to explain at real-world scale.
MC-19	b	The model links words to labels. 'free' appeared only in scam examples. Adding honest messages containing 'free' and retraining usually fixes this.
MC-20	b	The criteria are explicit and every decision must be explained; rules can name the criterion that failed.
MC-21	a	1956 → 1973 → 1980 → 1987.
MC-22	a, b, c	a–c are the causes given in Stop 4. Boolean logic dates from 1847 and is the foundation of computing, not a cause.
MC-23	b	5 is not 'less than 5' and not 'more than 5': a gap. Fix F2 to 'at least 5 and at most 10'.
MC-24	b	In Wander only G1 applies (player seen → Attack). G2 (low health → Flee) needs the state to be Attack, so fleeing comes one moment later.

C · Open-ended: key points and sample answers

Award credit for any well-reasoned answer that covers most key points; sample answers show one good response, and other valid examples are accepted.

OE-1 Expected key points *[S1 · Mission 1.3]*

- Truth table: T < 18 gives ON, OFF, OFF; T <= 18 gives ON, ON, OFF for 17, 18, 19 °C.
- The rules differ only at the boundary (exactly 18 °C).
- A computer follows the condition exactly; it cannot guess intent.

- A sensible consequence: e.g. heater runs more (energy cost), or in a safety system (medicine fridge alarm) a boundary error could trigger or miss an alarm.
- Good practice: state boundaries explicitly and test them (boundary testing).

Sample answer: The two rules are not the same. For 17 °C both give ON; for 19 °C both give OFF; but at exactly 18 °C, 18 < 18 is false (heater OFF) while 18 <= 18 is true (heater ON). A computer does exactly what the condition says and cannot work out what the programmer 'meant', so this one character changes the behaviour whenever the reading is exactly 18. In a thermostat this changes energy use; in a vaccine fridge alarm using 'T > 8', confusing > with >= could fail to warn at exactly 8 °C. That is why requirements must say precisely what happens at the boundary, and why testers deliberately try boundary values.

OE-2 Expected key points *[S2 · Missions 2.1, 2.3]*

- Cycle 1: R1 fires → is a bird. Cycle 2: R3 fires → is a penguin. Cycle 3: no rule adds a new fact → stop.
- R3 could not fire in cycle 1 because 'is a bird' was not yet known.
- Stop condition: a full cycle adds nothing new.
- Chain: penguin because R3 (bird + cannot fly + swims); bird because R1 (has feathers); remaining facts given by user.
- Trust: an expert can check each step and spot a wrong rule or fact; MYCIN could explain its reasoning on request.

Sample answer: Cycle 1 starts with has feathers, cannot fly, swims; R1 is the first rule whose IF part is true, so it fires and adds 'is a bird'. In cycle 2, R1 would add nothing new, R2 is false, and R3 (is a bird AND cannot fly AND swims) is now fully true, so it fires and adds 'is a penguin'. In cycle 3 no rule can add a new fact, so the engine stops. Asked 'why penguin?', the system answers: because of R3; 'is a bird' because of R1 from 'has feathers'; 'cannot fly' and 'swims' were given. A doctor using MYCIN could read such a chain, check every rule and fact, and reject the advice if any step was wrong, which is much easier to trust than an answer with no reasons.

OE-3 Expected key points *[S2 · Lab 2A, Mission 2.1]*

- R2 makes it a mammal; R5 (mammal, eats meat, black stripes) and R6 (mammal, hooves, black stripes) both match.
- Prolog/engine behaved correctly; the knowledge base has a rule conflict (overlapping conditions).
- Valid fix examples: add a 'does not have hooves' condition to R5 (e.g. a fact cannot_have hooves or not-has-hooves); or a rule rejecting carnivores with hooves; or ask the expert which is correct.
- Testing: rerun all earlier cases (pingu, hoot, stripes, tweety) plus the mystery case and compare with expected results (regression testing).
- Link: conflicts like this made large expert systems expensive to maintain (Stop 4).

Sample answer: R2 fires because the animal gives milk, so it is a mammal. Then R5 matches (mammal, eats meat, black stripes) and R6 matches too (mammal, hooves, black stripes), so Prolog proves both tiger and zebra. Prolog did exactly what the rules allow; the problem is a rule conflict in the knowledge base, because the two rules overlap for striped mammals. One fix is to make R5 require that the animal does not have hooves, for example is_a(X, tiger) :- is_a(X, mammal), fact(X, eats_meat), fact(X, has_black_stripes), \+ fact(X, has_hooves). Now only R6 matches the mystery animal. I would then re-run every earlier query (pingu, hoot, stripes, tweety) and check the answers are unchanged, because each new rule can create new conflicts; this is why large rule bases became costly to maintain.

OE-4 Expected key points *[S2 · Lab 2B]*

- ELIZA (Weizenbaum 1966) uses a script: keywords, decomposition patterns and reassembly templates; swaps pronouns.
- (1) keyword 'computer' → a computer/machine template.
- (2) no keyword → reuses a phrase stored earlier from a 'my …' sentence; inserting it into a template breaks grammar.
- (3) 'what' question keyword → deflecting template; no arithmetic is done.
- Demonstrates: rules over symbols can create an illusion of conversation. Does not demonstrate: understanding, reasoning, factual knowledge or meaning; links to 'the computer does not know what symbols mean'.

Sample answer: ELIZA works from a script of rules. It scans the input for a keyword, splits the sentence using a matching pattern, and builds a reply from a template, swapping pronouns such as my → your. In (1) the keyword 'computers' triggered a template about machines. In (2) there was no keyword, so ELIZA used a phrase it had stored earlier from 'my exams' and pasted it into a template, which is why the grammar is broken. In (3) the question word

'what' triggered a deflecting reply; ELIZA never calculated 2 + 2. These examples show that rules applied to symbols can look like conversation, which surprised people in 1966. But ELIZA has no idea what any word means, cannot answer factual questions, and does not reason, just as BEK's rules do not know what a feather is.

OE-5 Expected key points *[S3 · Mission 3.1, Lab 3, Mission 3.2]*

- At least three criteria: where knowledge comes from; data needed; explainability; handling unusual cases; updating/maintenance.
- Evidence: BEK's 'FREE MONEY' rule beaten by 'FR3E M0NEY'; To'ti's threshold model still wrong on F; Lab 3 errors on 'free'/'parcel' words; retraining after adding examples.
- ML: flexible, retrainable, needs labelled data, harder to explain, can be confidently wrong.
- Rules: explainable, predictable, but brittle and need constant hand-editing.
- Reasoned recommendation, e.g. ML for detecting patterns + rules for trusted contacts/blocked senders and explainable decisions.

Sample answer: A rule-based filter gets its knowledge from people who write rules such as 'IF message contains FREE MONEY THEN scam'. It is easy to explain, but brittle: scammers changed the spelling to 'FR3E M0NEY' and the rule stopped working, so someone must keep editing rules. A machine-learning filter learns from labelled messages; it can find patterns nobody wrote down and can be retrained when scams change. But it needs plenty of good labelled data, it is harder to explain, and it makes mistakes: our threshold model still misclassified message F, and in Lab 3 honest messages containing 'free' or 'parcel' could be called scams until we added better examples. I recommend a combination: a trained model scores each message, while explicit rules handle cases that must be predictable, such as never blocking a trusted contact, and give merchants a clear reason when a message is blocked.

OE-6 Expected key points *[S3 · Mission 3.1 Reflection]*

- The 1/10 was measured on the same data used to choose k: the model was fitted to it, so the estimate is optimistic.
- 10 examples are far too few to estimate accuracy reliably.
- Future messages may differ (new scam styles, or a genuine newsletter with many links).
- Test data = labelled examples not used in training; measure accuracy there.
- Possibly mention: more data, keep retesting over time.

Sample answer: The claim is weak. The model chose k = 4 precisely because it made the fewest mistakes on those ten messages, so measuring accuracy on the same messages is like marking your own exam after seeing the answers: it is optimistic. Training data is used to choose the model; test data is a separate set of labelled examples the model has never seen, used only to measure how well it works. Second, ten messages are far too few: one extra message such as N changed the best rule completely. Third, future messages may not look like the training set; a genuine university newsletter with twelve links would be called scam. A better method is to collect many more labelled messages, keep some aside as a test set, train only on the rest, and report accuracy on the test set, repeating this when new kinds of message appear.

OE-7 Expected key points *[S4 · Real thing, Missions 4.2, 4.4]*

- Expectations: 1980s boom; expert systems and Japan's Fifth Generation project (1982–1992) promised far more than delivered.
- Technology: brittle (gaps, e.g. Tweety/gives milk + swims), rule conflicts (mystery animal), no learning; LISP-machine market collapse 1987; Strategic Computing cut 1988.
- Combinatorial explosion: as rules grow, possible interactions/conflicts grow very fast, so testing and maintaining every combination becomes impractical (like adding digits to a lock).
- Clear cause → effect: funding and interest fell.
- Accurate dates/terms.

Sample answer: In the 1980s, companies and governments expected expert systems, and projects such as Japan's Fifth Generation programme (1982–1992), to deliver far more than they did: that is the expectations cause. In practice, rule-based systems had the weaknesses we saw ourselves at Stop 2. They were brittle: the animal identifier said only 'is a bird' for Tweety and 'is a mammal' for an animal that swims, because no rule covered those cases. And their rules could conflict: the mystery animal came out as both a tiger and a zebra. Real systems had thousands of rules, and the number of ways rules can interact grows explosively, just as each extra digit multiplies the codes on a lock by ten, so finding gaps and conflicts became very expensive. At the same time cheaper ordinary computers replaced special LISP machines, whose market collapsed in 1987: the technology cause. With results falling short, funding was cut and interest fell.

OE-8 Expected key points *[S5 · Lab 5]*

- Prolog answers goals backwards (backward chaining) and tries rules in order.
- Broken rule: to prove inside(X,Z) its first step is to prove inside(X,Y), the same kind of goal, without taking any real step, so it keeps calling itself until memory runs out (infinite recursion).
- Working rule: always takes one real located_in step first, moving one link along a finite chain, so it must reach the end.
- The knowledge is the same; only the rule's form/order changed: correctness depends on how rules are written, not just what they mean.
- Lesson: test rules, including 'no' answers (mongolia), and read error messages; ties to maintenance difficulty.

Sample answer: Prolog proves a goal by working backwards through the rules. With the broken rule, proving inside(registan, mongolia) first requires proving inside(registan, Y), which uses the same rule again, which again first requires another inside goal, and so on. The rule never forces Prolog to take a real step along a located_in link, so it keeps calling itself until it runs out of memory; SWISH reports this as a stack limit and 'probable infinite recursion'. The working rule always takes one real located_in step first, so each call moves one link along a finite chain (Registan → Samarkand → Uzbekistan → Central Asia) and must eventually succeed or fail; for Mongolia it simply answers false. Both rules 'mean' the same thing to a human, but only one works, which shows that rule-based systems must be tested with yes and no cases.

Review record (complete before submission)

The brief requires another member to attempt and review every question.

Part reviewed	Reviewer (did not write it)	Date	Problems found / changes made
A · True/False			
B · Multiple choice			
C · Open-ended			

AI assistance: drafted with AI assistance (Claude, Anthropic) and checked against the guide and programs; see docs/ai-use-log.md.